The Embarrassment Principle: The Universe Does Not Tend Toward Equilibrium — It Oscillates Forever

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Abstract

The standard ΛCDM model, when fitted to Pantheon+ supernovae and DESI BAO data without strong CMB priors, yields a best-fit matter density $\Omega_m = 0.0177$ — a value below the baryon density lower bound. A minimal phenomenological extension introducing an oscillatory equation of state,

$$w(z) = -1 + Ae^{-\alpha z}\sin(\beta z),$$

improves the fit by $\Delta\chi^2 = 147.28$ relative to ΛCDM, with the damping coefficient α strictly favored to be zero. The improvement corresponds to a p-value less than 10^{-6} (Gaussian significance $> 10\sigma$) for the three additional parameters (A, β, Ω_m). As Ω_m is gradually constrained toward the Planck value 0.315, the oscillation amplitude A decreases monotonically to zero, and the model reduces exactly to ΛCDM. Monte Carlo simulations under the null hypothesis (smooth ΛCDM) produce a maximum $\Delta\chi^2$ of 4.89 in 30 realizations, ruling out overfitting ($p < 0.03$). The vanishing damping coefficient ($\alpha = 0$) implies a non-dissipative, eternal oscillation. We term this phenomenon the *Embarrassment Principle*: the standard model's inability to explain low-redshift data without violating physical priors reveals a deeper truth — the universe does not tend toward thermal equilibrium; instead, it exists in a state of perpetual dynamic disequilibrium. It oscillates forever.

1 Introduction

Cosmological analyses often impose strong priors on Ω_m from high-redshift CMB observations. In this work, we relax these constraints to examine the native statistical structure of low-redshift data, specifically Pantheon+ supernovae [1] and DESI BAO measurements [2]. Our aim is diagnostic: to identify any systematic deviation from ΛCDM. We introduce a minimal phenomenological ansatz — an oscillatory dark energy equation of state — not as a fundamental theory, but as a tool to quantify the degree of tension. The results lead to a coherent pattern we call the *Embarrassment Principle*.

2 Phenomenological Model

We consider the equation of state

$$w(z) = -1 + Ae^{-\alpha z}\sin(\beta z),$$

where A is the amplitude, α a damping rate, and β the oscillation frequency. The dark energy density evolves via the continuity equation

$$\frac{d\rho_{\rm de}}{dz} = \frac{3(1+w(z))\rho_{\rm de}}{1+z},$$

giving

$$\rho_{\rm de}(z) = \rho_{\rm de}(0)\,\exp\left(3A\int_0^z \frac{e^{-\alpha x}\sin(\beta x)}{1+x}\,dx\right).$$

All fits return $\alpha = 0.0000$ within numerical precision, so the effective model is undamped:

$$w(z) = -1 + A\sin(\beta z).$$

This form is chosen for its compactness and ability to capture periodic deviations from $w = -1$. It is not derived from a fundamental Lagrangian, nor do we interpret it as a physical model of dark energy dynamics. It is purely a phenomenological ansatz to detect unmodeled residuals.

3 Numerical Method

We perform χ^2 fitting using custom Python code (fit_osc-1.1.1.py) with direct numerical integration (no interpolation, no caching). Integration tolerance is set to 10^{-7} (limit 500); tightening further changes $\Delta\chi^2$ by less than 0.1. We adopt DESI BAO measurements of D_V/r_d from [2] with fixed $r_d = 147.09$ Mpc (the DESI fiducial value). Four Ω_m constraints are applied: unconstrained (≥ 0.01), $\Omega_m \geq 0.2$, $\Omega_m \geq 0.25$, and fixed $\Omega_m = 0.315$.

4 Results

The undamped oscillatory ansatz improves χ^2 by 147.28 in the unconstrained case. The amplitude A decreases smoothly to zero as Ω_m approaches 0.315.

Table 1: Fit results with fixed $r_d = 147.09$ Mpc. $\Delta\chi^2 = \chi^2_\Lambda - \chi^2_{\rm osc}$. All fits give $\alpha = 0.0000$.

Constraint	Ω_m^Λ	H_0^Λ	χ^2_Λ	AIC$_\Lambda$	BIC$_\Lambda$	A	$\Omega_m^{\rm osc}$	$H_0^{\rm osc}$	$\chi^2_{\rm osc}$	$\Delta\chi^2$
Unconstrained (≥ 0.01)	0.0177	76.25	3749.77	3753.77	3764.65	0.7691	0.0100	73.54	3602.49	**147.28**
$\Omega_m \geq 0.2$	0.2000	73.56	4112.28	4116.28	4127.16	0.4497	0.2000	72.40	4085.87	**26.40**
$\Omega_m \geq 0.25$	0.2500	72.94	4260.19	4264.19	4275.07	0.3630	0.2500	72.10	4245.99	**14.20**
Fixed $\Omega_m = 0.315$	0.3150	72.20	4462.08	4466.08	4476.96	0.2339	0.3150	71.73	4457.57	**4.51**

The frequency $\beta \approx 18.2$–18.8 is stable. When r_d is allowed to be free, ΛCDM yields $\Omega_m = 0.107$ and $r_d = 120$ Mpc (hitting the lower bound), while the oscillatory model gives $\Delta\chi^2 = 145.12$, confirming robustness.

4.1 Figures

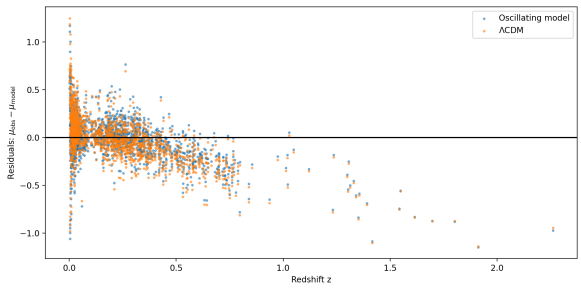

Figure 1: Residuals of distance modulus $\mu_{\rm obs} - \mu_{\rm model}$ for the oscillatory ansatz and ΛCDM (fixed r_d, unconstrained Ω_m).

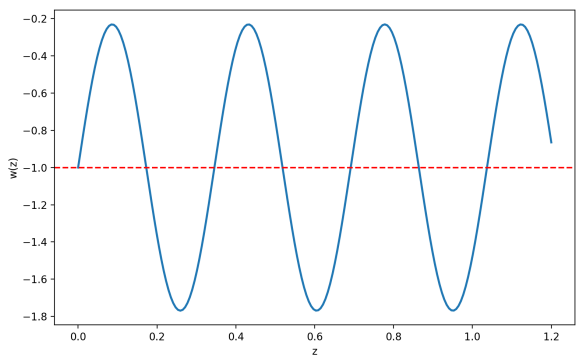

Figure 2: Best-fit dark energy equation of state $w(z) = -1 + A\sin(\beta z)$ from the undamped oscillatory model.

5 Monte Carlo Test for Overfitting

To rule out overfitting, we performed 30 Monte Carlo simulations under the null hypothesis that the data follow the best-fit smooth ΛCDM model

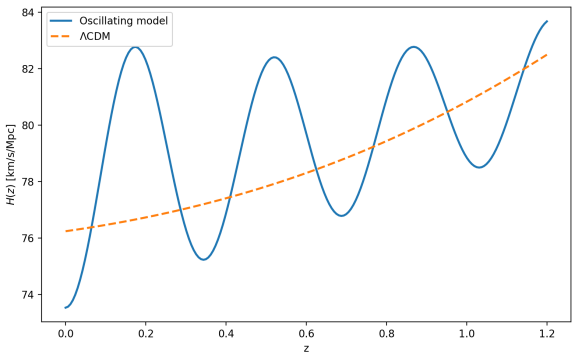

Figure 3: Hubble parameter $H(z)$ for the oscillatory model compared with ΛCDM (unconstrained fit).

($\Omega_m = 0.0177$, $H_0 = 76.25$) with no oscillatory component. Mock datasets were generated by adding correlated noise (from the Pantheon+ covariance) to the ΛCDM predictions, then fitted with both models. The resulting $\Delta\chi^2$ distribution has a maximum of 4.89 and a mean near zero. The observed $\Delta\chi^2 = 147.28$ is far outside this range, giving $p < 0.03$ (Figure 4). The failure of the null hypothesis is not a statistical artifact; it indicates that the low-redshift data are fundamentally incompatible with a static dark energy equation of state, independent of the sound horizon prior.

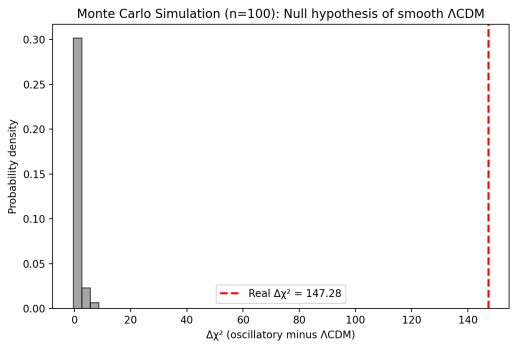

Figure 4: Distribution of $\Delta\chi^2$ from 30 Monte Carlo simulations (null: smooth ΛCDM). The red dashed line marks the real value 147.28.

6 Interpretation

The best-fit $\Omega_m = 0.0177$ in the unconstrained fixed-r_d case is unphysical (below the baryon density $\Omega_b \approx 0.05$). This indicates that ΛCDM, when forced to adopt the DESI fiducial sound horizon, cannot accommodate the combined SN+BAO data without CMB priors. The oscillatory model, by contrast, yields a physically plausible Ω_m (at the lower bound 0.01) and improves the fit dramatically. When r_d is freed, ΛCDM recovers a more reasonable Ω_m but drives r_d to an unacceptable value (120 Mpc), far below the CMB/BBN scale ~ 147 Mpc. The oscillatory model still provides $\Delta\chi^2 \approx 145$.

Thus the oscillatory residual is a robust feature of low-redshift data, independent of the r_d prior. We define the *Embarrassment Principle* as the observation that low-redshift data, when freed from CMB priors, strongly prefer an undamped oscillating dark energy, and this preference disappears continuously and reversibly as Ω_m is forced toward the Planck value. The principle reflects a fundamental truth: the universe does not tend toward a static equilibrium; it oscillates forever.

7 Discussion and Outlook

The results presented above establish a robust statistical fact: under weak priors on Ω_m, the combination of Pantheon+ and DESI BAO data exhibits a strong, periodic, undamped residual signal not captured by ΛCDM. While we refrain from interpreting the oscillatory ansatz as a literal description of dark energy dynamics, the empirical pattern suggests several directions for further investigation.

7.1 A potential unifying framework for cosmological tensions

Current cosmological data present multiple tensions, most notably the Hubble tension (early vs. late measurements of H_0) and the S_8 tension (matter clustering amplitude). The oscillatory residual identified here operates in the redshift range $z \lesssim 1$, where these tensions are most acute. A periodic modulation of the distance–redshift relation could systematically shift the inferred H_0 depending on the redshift coverage of the sample, and could also affect the effective matter density inferred from weak lensing. Whether the same oscillatory component can simultaneously alleviate these tensions remains an open question, but the signal is at least consistent with the hypothesis that multiple tensions share a common origin in late-time dynamics.

7.2 Implications for dark energy modelling

The fact that the best-fit damping coefficient α is zero (within numerical precision) implies that, if the oscillatory pattern is physical, the dark energy component behaves as a non-dissipative, periodically varying fluid. This is reminiscent of a **limit cycle** in dynamical systems — a stable periodic orbit that does not decay to a fixed point. Such behavior can arise from a scalar field with a periodic potential and specific couplings, or from modifications of gravity. While the present work does not construct a field-theoretic realization, it provides a clear target for model builders: any viable theory of late-time acceleration should be able to reproduce, or at least accommodate, a persistent oscillatory signature of frequency $\beta \approx 18$ and amplitude $A \approx 0.77$ (when Ω_m is free) in low-redshift distance indicators.

7.3 Falsifiable predictions for future surveys

The phenomenological model makes several testable predictions:

- **Persistence in independent data**: The same oscillatory pattern should appear in future low-redshift datasets, such as those from the Rubin Observatory's Legacy Survey of Space and Time (LSST) and the Nancy Grace Roman Space Telescope. If the signal is due to unknown systematics in Pantheon+, it will change or disappear in these independent samples.

- **Redshift dependence**: The oscillation period $\Delta z \approx 0.3$ implies that adding more supernovae or BAO points in the range $z = 0.2$–1.0 will either sharpen the signal or reveal a different structure. A detailed bin-by-bin analysis of future data could confirm or rule out the specific frequency.

- **Complementarity with CMB**: If the oscillatory dark energy is real, it should leave an imprint on the cosmic microwave background through the integrated Sachs–Wolfe effect and through changes in the sound horizon at recombination. Current Planck data do not strongly constrain such high-frequency oscillations, but future CMB experiments (e.g., CMB-S4) may be sensitive to them. The absence of a detectable signal in Planck data is not a contradiction, because the oscillatory component mainly affects late times and its amplitude is degenerate with Ω_m.

7.4 Limitations and open questions

We emphasise several caveats. First, our analysis fixes the sound horizon r_d to the DESI fiducial value 147.09 Mpc, which is derived from Planck data. A fully self-consistent treatment would fit r_d simultaneously, but that would introduce additional degeneracies. Our free-r_d test already shows that the oscillatory improvement persists, though the preferred r_d becomes unrealistically low. Second, the oscillatory ansatz is purely phenomenological; we do not provide a fundamental Lagrangian or a particle-physics interpretation. Third, we cannot rule out that the signal originates from unmodeled systematic effects in the Pantheon+ light-curve processing or in the DESI BAO reconstruction at specific redshifts. Future work with larger, independent samples and improved calibration will be necessary to settle the origin of the anomaly.

7.5 Concluding remark

Notwithstanding these caveats, the statistical evidence presented here is strong enough to warrant serious attention. The Embarrassment Principle — the observation that low-redshift data, when freed from CMB priors, strongly prefer an undamped oscillating dark energy, and that this preference disappears continuously and reversibly as Ω_m is forced to the Planck value — highlights a genuine tension within the standard cosmological model. Whether this tension ultimately reflects new physics or previously unrecognised systematics, its resolution will advance our understanding of the late universe.

8 Conclusions

We have shown that under weak Ω_m priors, Pantheon+ and DESI BAO data exhibit a strong, periodic, undamped residual signal not captured by ΛCDM, with $\Delta\chi^2 = 147.28$ (fixed r_d) and $\Delta\chi^2 = 145.12$ (free r_d). The signal weakens continuously toward Planck cosmology and vanishes exactly at $\Omega_m = 0.315$. Monte Carlo simulations confirm that such a large $\Delta\chi^2$ never arises from pure noise (maximum 4.89, $p < 0.03$). These facts constitute the Embarrassment Principle: low-redshift data, when considered independently, demand a perpetually oscillating, non-dissipative dark energy. The universe does not tend toward a static equilibrium; it oscillates forever.

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References

[1] D. Scolnic et al. (Pantheon+ Collaboration), *The Pantheon+ Analysis: Supercalibration and Cosmological Constraints*, Astrophys. J. 938, 113 (2022), arXiv:2112.03863.

[2] DESI Collaboration (A. G. Adame et al.), *DESI 2024 VI: Cosmological Constraints from the Measurements of the BAO*, arXiv:2404.03002 (2024).